

INTEGRATED WIRING GUIDE, BOM & TECHNICAL LAYOUT MANUAL

Complete Companion Deployment Document • DWG. NO. LR-ELEC-001

This master technical manual fully integrates the step-by-step assembly workflows, structured Bill of Materials, and the four specialized reference layouts for the interior electronics of the vintage L&R Master base casting. Refer to these integrated resources side-by-side once the VHT wrinkle paint finish completes its current cure window.

1. Harness Wire Bill of Materials (BOM) & Pre-Cut Specs

Pre-measure, cut, and tin your wire lines before committing them to the tight interior chassis wells to prevent accidentally brushing a hot soldering iron tip against fresh exterior casing paint surfaces.

From Connection	To Connection	Color	Gauge / Type	Est. Len.	Shrink Tube
P1 Main Cord (Hot)	S1 Plunger Button (Line-In Terminal)	Black	16 AWG Silicone	5.5 inches	3/16" High-Temp
S1 Plunger Button	LMP1 Lamp Terminal & R1 Resistor Input	Black	16 AWG Silicone	3.5 inches	1/8" High-Temp
P1 Main Cord (Neu)	RH1 Rheostat Base (Terminal Block Log)	White	16 AWG Silicone	6.0 inches	3/16" High-Temp
S2 Toggle Switch	RH1 Rheostat Wiper Swing Terminal	White	16 AWG Silicone	4.0 inches	1/8" High-Temp
R1 Dropping Resistor	S2 Toggle Switch (Input Terminal)	Black Wire	16 AWG Silicone	3.0 inches	1/8" High-Temp
P1 Main Cord (Gnd)	Chassis Casting Screw (Bare Alum Boss)	Green	16 AWG Stranded	4.5 inches	Ring Lug Ins.
Upper Chassis Column	Motor Neck Strain Relief Wire Exit Port	BLK/WHT/GRN	Original 3-Wire	Harness Tail	Dual Wall Glue

**RESTORATION CRITICAL WARNING: THE "GREEN NECK WIRE" EXCEPTION**

As documented during diagnostic teardown profiling, the integrated vintage ****Green neck line**** leading directly down out of the upper motor housing profile is an active ****LIVE current-carrying extension**** of the series motor loop circuit net. Under operating voltage loads, it remains active. Treat it strictly as a functional circuit conductor as dictated in your schematic layout—****do not connect this green motor neck wire to the aluminum chassis frame.****

2. Step-by-Step Electrical Routing & Assembly Sequence

Phase I: Protective Chassis Earth Ground Integration

Locate a non-visible structural internal casting boss screw point away from the rotating sweep space of the

center rheostat knob mechanism. Use an abrasive tool to scrape clear a microscopic bright silver circle of bare aluminum, entirely piercing through the interior overspray layer of VHT paint. Crimp your incoming wall-plug green safety ground lead securely into an insulated ring terminal, applying a star washer directly beneath the screw head to bite deep into the chassis body, establishing an armored protective grounding path.

Phase II: Power Entrance & Control Switch Node Solder Operations

Fish your fresh 110V utility line power supply cord into the lower base opening, anchoring it via a heavy-duty strain relief grommet. Route the main incoming Black Hot wire line straight to the bottom leaf node terminal configuration of the momentary plunger push-button switch. From the opposite terminal side of the plunger button, run a jumper down to bridge across the input anchor of the power drop ceramic resistor tube, keeping paths tight to maintain clean spacing beneath the indicator lamp housing.

Phase III: Integrating the Heater & Rheostat Loops

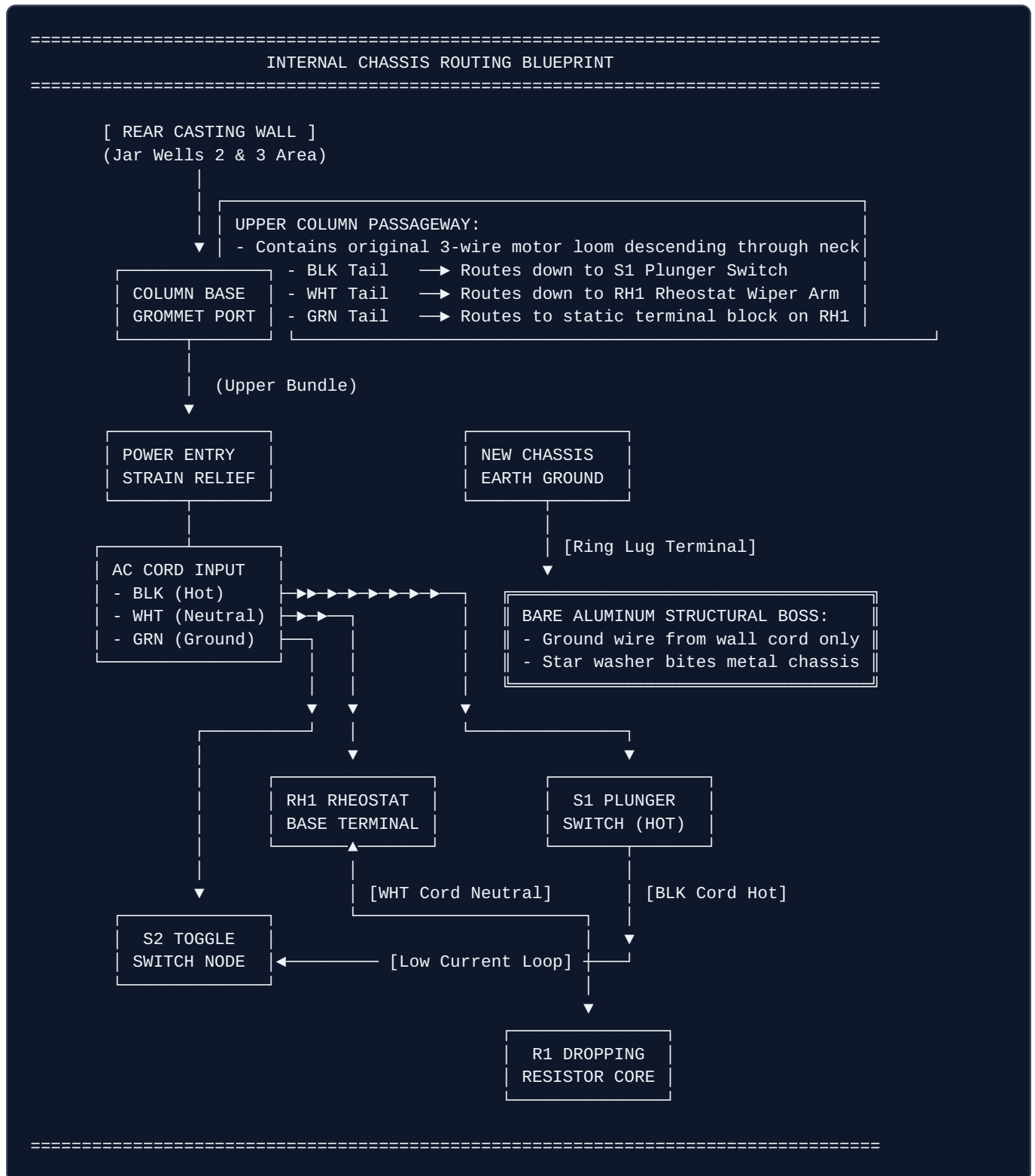
Solder the output leg of the dropping power resistor straight into the rear contact pole of the toggle switch. Take the White output return line coming away from the toggle switch and hook it into the low-current sweep side of your cleaned open-coil rheostat wiper arm. Loop your incoming AC Neutral (White line) straight to the static terminus log block located at the base platform array of the rheostat to lock in your variable series speed loop architecture.

Phase IV: Upper Column Neck Tail Integration

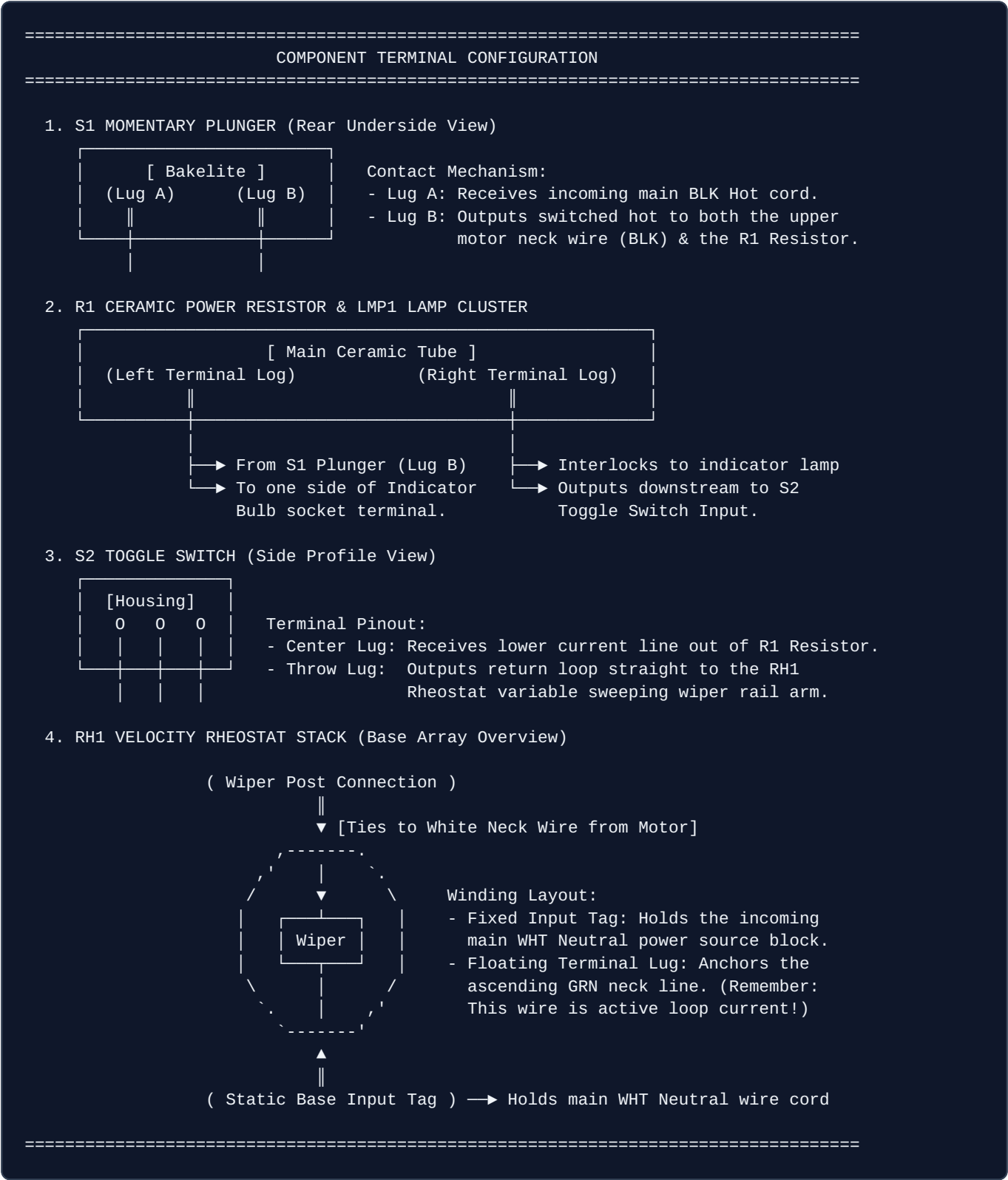
Feed the pre-sorted three-wire harness descending from the chrome pillar column into the interior cavity. Solder the designated loop nodes cleanly to their landing spots on the rheostat terminal board tags according to your schematic's 7-terminal layout. Slip clean, high-visibility heat-shrink sleeving completely over all bare copper splice junctions, using your heat gun to shrink the tubing uniformly into place without letting it touch the nearby freshly cured casing wall paint.

3. Integrated Technical Layouts

Layout 1: Internal Chassis Routing Path Map



Layout 2: Physical Terminal Connection Map (Component Level)



Layout 3: Harness Bill of Materials (BOM) Bench-Card

HARNESS WIRE SPECIFICATION CARD				
[ID]	[COLOR]	[GAUGE / TYPE]	[CUT LENGTH]	[TERMINATION DESIGNATION]
W-1	BLACK	16 AWG Silicone	5.5 in.	AC Line Inlet → Plunger S1
W-2	BLACK	16 AWG Silicone	3.5 in.	Plunger S1 → Resistor R1
W-3	WHITE	16 AWG Silicone	6.0 in.	AC Line Inlet → Rheostat base
W-4	WHITE	16 AWG Silicone	4.0 in.	Toggle S2 → Rheostat wiper
W-5	BLACK	16 AWG Silicone	3.0 in.	Resistor R1 → Toggle S2
W-6	GREEN	16 AWG Stranded	4.5 in.	AC Plug Gnd → Casting Chassis
[INSULATION PROTECTION CRITERIA]				
* Use 125°C or higher rated cross-linked polymers (Silicone/PTFE rubber sleeve).				
* Avoid using standard hardware-store PVC primary wire next to the resistor well.				

Layout 4: Post-Assembly Cold-Check Diagnostic Matrix

UN-POWERED BENCH CHECKLIST MATRIX	
STEP 1: SAFETY CASING SHIELDING CHECK	
	Probe 1: Round Earth Pin on 3-Prong AC Wall Plug. Probe 2: Raw Exposed Bare Metal Screw on Casing Frame. TARGET VALUE: Less than 0.2 Ohms ($\leq 0.2 \Omega$) → Confirms solid ground fault path.
STEP 2: HIGH-VOLTAGE HOT CHASSIS SHORT CHECK	
	Probe 1: Flat Hot Blade Pin on AC Wall Plug. Probe 2: Chrome Upper Column Guide Pillar. TARGET VALUE: Open Loop / Infinite Resistance (O.L.) → Confirms no pinch shorts.
STEP 3: HIGH-VOLTAGE NEUTRAL CHASSIS SHORT CHECK	
	Probe 1: Flat Neutral Blade Pin on AC Wall Plug. Probe 2: Chrome Upper Column Guide Pillar. TARGET VALUE: Open Loop / Infinite Resistance (O.L.) → Confirms no loop leaks.
STEP 4: FUNCTIONAL CIRCUIT CONTINUITY CHECK	
	Probe 1: Flat Hot Blade Pin on AC Plug. Probe 2: Flat Neutral Blade Pin on AC Plug. Action: Actuate and depress the top mechanical Plunger Button switch. TARGET VALUE: ~28 Ohms plus the sweeping rheostat coil resistance gradient.